R Code

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###################### Control Structures in R ####################
#### Loops ###########

# Repetitive commands are executed by loops

#For loop
#While loop
#repeat loop

#For loop

# 1# Print 1 to 10
for(i in 1:10){
  print(i)
}

#2# Sum of First 10 Natural Numbers
sum_value <- 0

for(i in 1:10){
  sum_value <- sum_value + i
}

print(sum_value)

#3
for(i in 15:20){
  print(i^2 + i^3)
}

#4# Factorial
n <- 5
fact <- 1

for(i in 1:n){
  fact <- fact * i
}

print(fact)

#5# Loop Over Vector
x <- c(2,4,6,8)

for(value in x){
  print(value^2)
}

#6# Generating a Multiplication Table (Matrix Form)
n <- 5
table_matrix <- matrix(0, n, n)
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for(i in 1:n){
  for(j in 1:n){
    table_matrix[i,j] <- i*j
  }
}

print(table_matrix)

#7# Row-wise Sum of a Matrix
A <- matrix(1:12, nrow=3)

row_sum <- numeric(nrow(A))

for(i in 1:nrow(A)){
  row_sum[i] <- sum(A[i, ])
}

print(row_sum)

#8# Column-wise Mean
A <- matrix(1:12, nrow=3)

col_mean <- numeric(ncol(A))

for(j in 1:ncol(A)){
  col_mean[j] <- mean(A[, j])
}

print(col_mean)

#9# Check Prime Numbers
num <- 29
is_prime <- TRUE

for(i in 2:(num-1)){
  if(num %% i == 0){
    is_prime <- FALSE
    break
  }
}

print(is_prime)

#10# Approximate Definite Integral Using Riemann Sum

# Function definition
f <- function(x){
  return(x^2)
}

# Interval
a <- 0
b <- 1
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# Number of subintervals
n <- 1000

h <- (b - a)/n
sum_value <- 0

for(i in 0:(n-1)){

  x_i <- a + i*h
  sum_value <- sum_value + f(x_i)
}

integral_approx <- h * sum_value

cat("Approximate value:", integral_approx, "\n")
cat("Exact value:", 1/3)

#11# Find the Maximum Element in a Vector

# Given vector
x <- c(12, 45, 7, 89, 23, 56)

# Assume first element is maximum
max_value <- x[1]

# Loop through remaining elements
for(i in 2:length(x)){

  if(x[i] > max_value){
    max_value <- x[i]
  }
}

print(max_value)

#######################  while loop ####################################

#Syntax
while(condition){
  # statements
}

#1 #Print Numbers from 1 to 5
i <- 1

while(i <= 5){
  print(i)
  i <- i + 1
}

#2# Print Even Numbers up to 10

i <- 2
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while(i <= 10){
  print(i)
  i <- i + 2
}

#3 Sum of First 10 Natural Numbers

i <- 1
sum_value <- 0

while(i <= 10){
  sum_value <- sum_value + i
  i <- i + 1
}

print(sum_value)

#4# Print Even Numbers Less Than 10

i <- 2

while(i < 10){
  print(i)
  i <- i + 2
}

#5 Print Cubes of Odd Numbers Less Than 10

i <- 1

while(i < 10){
  print(i^3)
  i <- i + 2
}

#6 Multiply Until Product Exceeds 100

i <- 1
product <- 1

while(product <= 100){
  product <- product * i
  i <- i + 1
}

print(product)

#7 Find Smallest n Such That n^2 > 200

n <- 1

while(n^2 <= 200){
  n <- n + 1
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}

print(n)

#8 Check if a Number is Prime
num <- 29
i <- 2
is_prime <- TRUE

while(i <= num-1){
  if(num %% i == 0){
    is_prime <- FALSE
    break
  }
  i <- i + 1
}

print(is_prime)

############################## Repeat loop ##################################

#Syntax
repeat {
# statements
if(condition) break
}

#1# Print 1 to 5

i <- 1

repeat{
  print(i)
  i <- i + 1

  if(i > 5){
    break
  }
}

#2# Sum of Natural Numbers Until Sum Exceeds 20

i <- 1
sum_value <- 0

repeat{
  sum_value <- sum_value + i
  i <- i + 1

  if(sum_value > 20){
    break
  }
}

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print(sum_value)

#3#
i <- 1

repeat{
  print(i^2)
  i <- i + 2
  if(i > 10) break
}
#4# Print Cubes of Odd Numbers Less Than 15
i <- 1

repeat{
  print(i^3)
  i <- i + 2

  if(i > 15) break
}

#5# Print Even Numbers and Stop After 20
i <- 2

repeat{
  print(i)
  i <- i + 2

  if(i > 20) break
}

#6# Print Multiples of 5 Until 50
i <- 5

repeat{
  print(i)
  i <- i + 5

  if(i > 50) break
}
#7# Print Squares Starting from 2 Until Square Exceeds 100

i <- 2

repeat{
  print(i^2)
  i <- i + 1

  if(i^2 > 100) break
}



###################### Practice Questions ################################

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######################## Practice Set 1 (For loop)

#Q1. Print the values of 3i+2 for i = 1 to 12.
#Q2. Given the vector x = c(4, 9, 16, 25, 36). Print the square root of each element.
#Q3. Given a vector x = c(10, 25, 7, 18, 42, 5). Find the minimum element using for loop.
#Q4. Print all numbers between 1 and 50 that are multiples of 4 but not multiples of 8.
#Q5. Print the cubes of odd numbers from 1 to 15.
#Q6. Find the factorial of 6.
#Q7. Find the sum of all even numbers between 1 and 50.
#Q8. Find the largest number among c(14, 27, 35, 9, 42, 18).

########################## Practice Set 2 (while loop)##########################

#Q1. Print numbers from 1 to 10 using a while loop.
#Q2. Print even numbers less than 20.
#Q3. Find the smallest n such that n^2> 150.
#Q4. Compute the sum of first 10 natural numbers.
#Q5. Print numbers starting from 3 and keep adding 4 each time until the number exceeds 50
#Q6. Print cubes of numbers starting from 1 until the cube exceeds 300.
#Q7. Given a number 8642, compute the sum of its digits using a while loop.


######################## Practice Set 3 (repeat loop)##########################

#Q1. Print numbers from 1 to 8 using a repeat loop.
#Q2. Print squares of numbers starting from 1 and stop when the square exceeds 50.
#Q3. Keep adding natural numbers (starting from 1) until the sum exceeds 40.
#Q4. print the sequence 3i-1 starting from i = 1 and stop when the value exceeds 50.
#Q5. compute the sum of consecutive even numbers starting from 2 and stop when the sum exceeds 150.
#Q6. compute the harmonic series 1 + 1/2 + 1/3 +......and stop when the sum exceeds 5.
#Q7. compute the binomial expansion terms of (1+x)^n for x=2 starting from n=1 and stop when the value exceeds 300.

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